cozaint®

MARCIA™ EBOOK · SECOND EDITION

# The Architecture of Intelligent Video Lifecycle Management

Why enterprise surveillance needs more than disk




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# About this edition

This book began in 2023 as *Video Surveillance Storage: Implementing a 2-Tier Storage Solution*. Its argument has not changed: surveillance video should be recorded at the quality the cameras were bought for, kept for as long as it is needed, and stored on the technology that fits each stage of its life, not crammed onto ever-larger racks of spinning disk.

What has changed is how practical that argument has become. The first edition described the ideal: an operator who simply moves the video management system (VMS) timeline and gets back any recording, whether it sits on disk or on LTO tape, with no extra steps and no IT staff. Cozaint has since built that workflow into **MARCIA™ Intelligent Video Lifecycle Management** and patented it (U.S. Patent No. 12,755,902).

This second edition keeps the educational core of the first: why disk-only storage becomes expensive, what LTO is, how a multi-tier architecture works, how to size it and what it costs. It adds what we have learned since: why retained video is becoming more valuable, how the patented workflow works, what the economics look like at enterprise scale, and how Hollywood's experience points the way.

**The idea in one sentence.** Media & Entertainment proved LTO for video; MARCIA makes it practical for surveillance; patented VMS integration makes the storage tier invisible to the operator. The result is substantially more retained video at dramatically lower lifecycle cost.

## How this book is organized

| Part | Chapters |
|---|---|
| The problem | 1. The video storage problem · 2. Why video is worth keeping |
| The technology | 3. LTO: not your grandfather's tape · 4. Why disk alone falls short |
| The architecture | 5. The architecture of Intelligent Video Lifecycle Management · 6. MARCIA and the patented workflow · 7. Hollywood comes to video surveillance |
| The decision | 8. What about the cloud? · 9. Sizing a system · 10. The economics · 11. Real-world examples |
| The implementation | 12. The physical architecture · 13. Best practices and getting started |

## Foreword to the first edition

*Written for the 2023 edition and reproduced as published.*

Welcome to this introduction of what I believe is a useful ebook on 2-tier video surveillance storage systems utilizing Linear Tape-Open (LTO) technology as the second tier. I'm Rich Gadomski, Head of Tape Evangelism at FUJIFILM, and I'm thrilled to be sharing with you some insights about this fascinating and increasingly relevant topic.

As you know, video surveillance has become an essential tool in many sectors of our society, from law enforcement to retail, transportation, and critical infrastructure. The ability to capture and analyze video footage has enabled organizations to enhance their security, prevent crimes, optimize operations, and make data-driven decisions. However, with the proliferation of high-definition cameras, the exponential growth of video data, and the increasing demand for longer retention periods, managing video surveillance storage has become a significant challenge for many IT and security departments.

That's where the concept of 2-tier video surveillance storage systems comes in. Rather than storing all video data on expensive, energy intensive, high-performance storage devices such as hard disk drives, organizations can leverage a 2-tiered approach that provides for the quick access of the most recent video and the availability of all recorded video no matter when it was originally stored. This approach can significantly reduce storage costs, optimize system performance, reduce carbon footprint and simplify video data management.

But what is LTO technology, and why is it a useful component of a 2-tier video surveillance storage system? Linear Tape-Open, or LTO, is an open-format magnetic tape technology that was first introduced in 2000 by IBM, HPE, and Quantum. Since then, LTO has become a widely adopted standard for data backup and archiving, thanks to its high capacity, fast transfer rates, reliability, low energy consumption and low cost per gigabyte. In a 2-tier video surveillance storage system, LTO serves as the second tier where older or less frequently accessed video data is replicated from the primary storage layer, such as hard disk arrays.

By using LTO tapes, organizations can store massive amounts of video data in a cost-effective and scalable manner, while also ensuring data integrity and long-term retention. LTO tapes can currently provide up to 18TB of capacity per

cartridge and transfer rates of up to 400MB/s, making them ideal for large-scale video surveillance environments. And these capacities and speeds are already set to increase.

But LTO technology is not just useful for its cost, capacity, energy profile and speed. LTO also offers advanced features that are critical for video surveillance storage, such as data encryption, WORM (Write-Once-Read-Many) functionality, and LTFS (Linear Tape File System) support. With data encryption, organizations can protect their video data from unauthorized access or theft, while WORM functionality ensures that once data is written to the tape, it cannot be overwritten or erased, thus preserving the authenticity and legal admissibility of the data. LTFS enables users to access, manage, and share video data on LTO tapes as if they were on a disk or a network drive, simplifying data access and reducing playback times.

In today's video surveillance infrastructure, where data volumes are growing exponentially, and regulatory and compliance requirements are becoming more stringent, 2-tier video surveillance storage systems utilizing LTO technology as the second tier are becoming very compelling. This book will delve deeper into the technical aspects of 2-tier video surveillance storage systems utilizing LTO technology, covering topics such as system architecture, data replication, LTO versus hard disk only offerings, LTFS and more. It will also provide practical tips and best practices for designing, implementing, and managing such systems, based on real-world scenarios and use cases.

Rich Gadomski, Head of Tape Evangelism, FUJIFILM Recording Media, USA, Inc. (2023)

# 1. The video storage problem

Video surveillance has become an essential part of modern security. High-quality recordings deter crime, speed up investigations, support prosecutions, protect workers and help organizations run better. Video management software (VMS) records, stores and manages that video, and for most of the industry's history the VMS has written everything to hard disk.

That approach is becoming more expensive every year, for reasons that compound one another:

- **More cameras.** Organizations keep adding them, and every camera adds video around the clock.
- **Higher resolution and frame rates.** A single 4K camera can generate up to 12 GB of video per hour. Higher quality is the reason cameras are bought, and it multiplies storage.
- **Longer retention.** Regulations, policies, litigation and investigations all push retention from days to months or years.
- **The cost of disk does not stop at purchase.** Hard drives spin and draw power 24 hours a day, every watt must be cooled, drives fail and wear out, and larger arrays need more enclosures, servers, rack space and support.

## The industry has been solving the wrong end of the problem

Faced with those costs, the surveillance industry has mostly changed the video instead of the storage:

- lowering camera **resolution**, which discards detail needed to identify people and objects;
- lowering **frame rate**, which loses fast motion;
- lowering **bit rate**, which degrades quality;
- **recording only on motion**, which misses events that the motion settings did not catch;
- and, when all else fails, **shortening retention**, which simply throws evidence away.

Each of these saves disk at the expense of the very thing the cameras were bought to capture. Organizations purchase high-definition cameras to obtain the best video available, and then record that video degraded or not at all.

> **The purpose of this book** is to remove those misguided methods of dealing with the video storage problem: record all video at the best quality, so as not to compromise safety, security or business value, and change the storage architecture instead.

## Why "single-tier" thinking persists

Most surveillance systems treat storage as one tier: whatever the VMS records goes to disk and stays there until it is overwritten. Other industries that manage large volumes of data stopped thinking this way long ago. They place data on the storage that matches how often it is used: fast, expensive storage for active data; dense, low-cost storage for everything else. Surveillance video, which is written once and rarely viewed after the first few days, is an ideal candidate for the same approach.

# 2. Why video is worth keeping

The first edition focused on cost: storage is expensive, and a multi-tier architecture makes it affordable. That is still true. But there is a bigger reason to rethink surveillance storage: **historical video is becoming more valuable.**

**Storage savings start the conversation. Preserving Trusted Visual Evidence is why it matters.**

## The clip that matters is often the one nobody knew mattered

| Reason to keep video | What happens without it |
|---|---|
| Investigations | Fraud, theft and misconduct are often discovered weeks or months after the fact |
| Litigation and liability | Slip-and-fall claims, disputes and lawsuits are frequently filed long after the event; without video, organizations cannot defend themselves |
| Compliance | Records rules, gaming regulations, corrections policies and evidence rules set minimum retention periods |
| Incident reconstruction | Understanding how an event unfolded often requires video from before anyone knew there was an event |
| AI re-analysis | New analytics can re-examine history for patterns, people, vehicles and events, but only if the history still exists |

## Quality matters as much as quantity

Retained video is only useful if it is good enough to use. The ability to identify a face, read a plate or see a card switched at a blackjack table depends on resolution and frame rate. Video kept at reduced quality to save disk may be present but useless.

## Chain of custody

Video submitted as evidence must be the original, unaltered recording, handled in a way that can be documented. An archive that keeps the original recordings and returns them through the same VMS used to record them supports that chain of custody far better than ad hoc copies and exports.

If video is an asset, the question is no longer “how little can we keep?” but “how can we afford to keep what we need, at full quality, and still get it back quickly?” The rest of this book answers that question.

# 3. LTO: not your grandfather's tape

Linear Tape-Open (LTO) is a high-capacity, reliable, energy-efficient and cost-effective digital data storage technology. It was introduced in 2000, is developed by Hewlett Packard Enterprise, IBM and Quantum through the LTO Program, and has become the industry standard for backup and archiving. It is also heavily used in Hollywood and broadcasting to archive video.




LTO: a palm-sized cartridge that holds 30 TB or more of video

## This is not VHS

When people in security hear "tape," many think of the VHS cassettes once used in analog surveillance systems. The two have almost nothing in common:

| | VHS (analog video tape) | LTO (digital data tape) |
|---|---|---|
| **Purpose** | Consumer analog video recording | Enterprise digital data storage and archiving |
| **Capacity** | Up to about 6 hours of analog video | 30 TB per LTO-10 cartridge, up to 40 TB with the newest cartridges |
| **Quality over time** | Degrades with age and each playback | Digital; error correction verifies data as it is written |
| **Recording** | Helical scan | Linear serpentine recording at up to hundreds of MB per second |
| **Lifespan** | Years | Media manufacturers rate LTO for up to 50 years of archival life |

## Why LTO fits surveillance video

Video is a **sequential** stream: each frame follows the one before it. LTO writes and reads data sequentially, one block after another, as the tape moves past the read/write head. That makes surveillance video and LTO a natural match: video is written to tape in the same order it was recorded and can be read back the same way.

Surveillance video also has the profile LTO was designed for:

- **vast amounts of data;**
- **rarely accessed after the first few days,** until it is needed;
- **sequential** data;
- a desire for **long-term retention.**

## The advantages of LTO

- **Scalability.** LTO has a published roadmap of generations with predictable capacity growth, extending to LTO-14 at a planned 576 TB per cartridge, more than half a petabyte on one palm-sized cartridge.
- **Reliability.** Built-in error correction and verification; modern LTO bit error rates are orders of magnitude better than hard disk.
- **Cost.** A very low cost per terabyte and a low total cost of ownership.
- **Durability.** Rugged cartridges protect the tape from dust and handling, and manufacturers rate the media for up to 50 years.
- **Energy.** A cartridge draws power only while it is being read or written. In a library slot or on a shelf it draws none and needs no cooling. Studies cited by the tape industry show tape systems using about 87% less energy than an equivalent capacity of hard disk over time, and producing about 97% less CO2e over their lifecycle.
- **Security.** Hardware encryption, write-once (WORM) cartridges, and the option to keep cartridges offline, out of reach of the network.

## LTFS: tape that describes itself

The Linear Tape File System (LTFS), introduced in 2010, is an open standard that lets a tape be read much like a removable drive. One partition holds the content and the other holds its index, so the cartridge is self-describing. Users can browse a tape, find files and read them directly without restoring the whole tape, on many operating systems and without proprietary software. For long-term archives, that openness matters: the archive is not locked to one server or one vendor.

As Carnegie Mellon professor James Bain put it: “The message about tape is that, just as SSDs did not displace HDDs, HDDs, in turn will not displace tape systems. All of them co-exist in a tiered storage architecture where cost and latency are balanced at every level of the system.”

## The LTO generations

| Generation | Year | Native capacity | Notable addition |
|---|---|---|---|
| LTO-1 | 2000 | 100 GB | |
| LTO-2 | 2003 | 200 GB | |
| LTO-3 | 2005 | 400 GB | WORM |
| LTO-4 | 2007 | 800 GB | Drive-level encryption |
| LTO-5 | 2010 | 1.5 TB | LTFS |
| LTO-6 | 2012 | 2.5 TB | Barium ferrite media |
| LTO-7 | 2015 | 6 TB | |
| LTO-8 | 2017 | 12 TB | |
| LTO-9 | 2021 | 18 TB | Up to 400 MB/s |
| LTO-10 | 2025 | 30 TB (up to 40 TB with the newest cartridges) | |
| LTO-11 to LTO-14 | Roadmap | Up to a planned 576 TB | |

## Tape libraries

An LTO library (sometimes called a “jukebox”) holds cartridges in slots and one or more tape drives, with a robot that moves cartridges between them. The library keeps track of where every cartridge is, so data can be retrieved without anyone touching a tape.

Libraries scale from a compact 2U unit with a single drive and a couple of dozen slots to enterprise libraries with dozens of drives and thousands of cartridges, holding many petabytes. Most grow by adding expansion modules, drives and cartridges, so a system can start small and grow with retention.

# 4. Why disk alone falls short

Hard disk drives (HDDs) are the right place for video that is being recorded and reviewed now. They are the wrong place to keep months or years of video that is rarely viewed.

## The costs that keep coming

- **Capacity costs scale with retention.** Double the retention and you double the disk.
- **Power and cooling.** Disks spin 24 hours a day, and every watt they draw must also be cooled.
- **Failures and wear-out.** Drives are mechanical. They fail at random during service and wear out after about five years of continuous use, so long-lived archives must replace every drive at least once.
- **RAID overhead, space and support.** RAID 6 typically uses two of every twelve drives for protection, and more drives mean more enclosures, servers, rack space and support contracts.

## The hard disk roadmap

Hard disk capacity has grown enormously since IBM's first 5 MB drive in 1956, and heat- and microwave-assisted recording (HAMR, MAMR) push it further. But as drives approach the physical limits of areal density, capacity growth is slower and costlier than on the LTO roadmap, and every additional terabyte still spins and draws power.

**Shingled magnetic recording (SMR)** drives raise capacity by overlapping tracks, but they are a poor fit for surveillance recording: writes are slower, RAID rebuilds are difficult and reads can bottleneck. SMR drives should be avoided in video surveillance recording systems.

## What about solid-state drives?

SSDs offer fast access and low latency, but for large-scale video retention they are the most expensive option per terabyte, have a finite number of write cycles that continuous recording consumes, and still draw power whenever they are online. They have a place in high-performance tiers, not in the long-term archive.

## The work-arounds, revisited

Because disk is costly, the industry has tried to make video smaller. None of these addresses the real issue, that disk is the wrong medium for long-term retention:

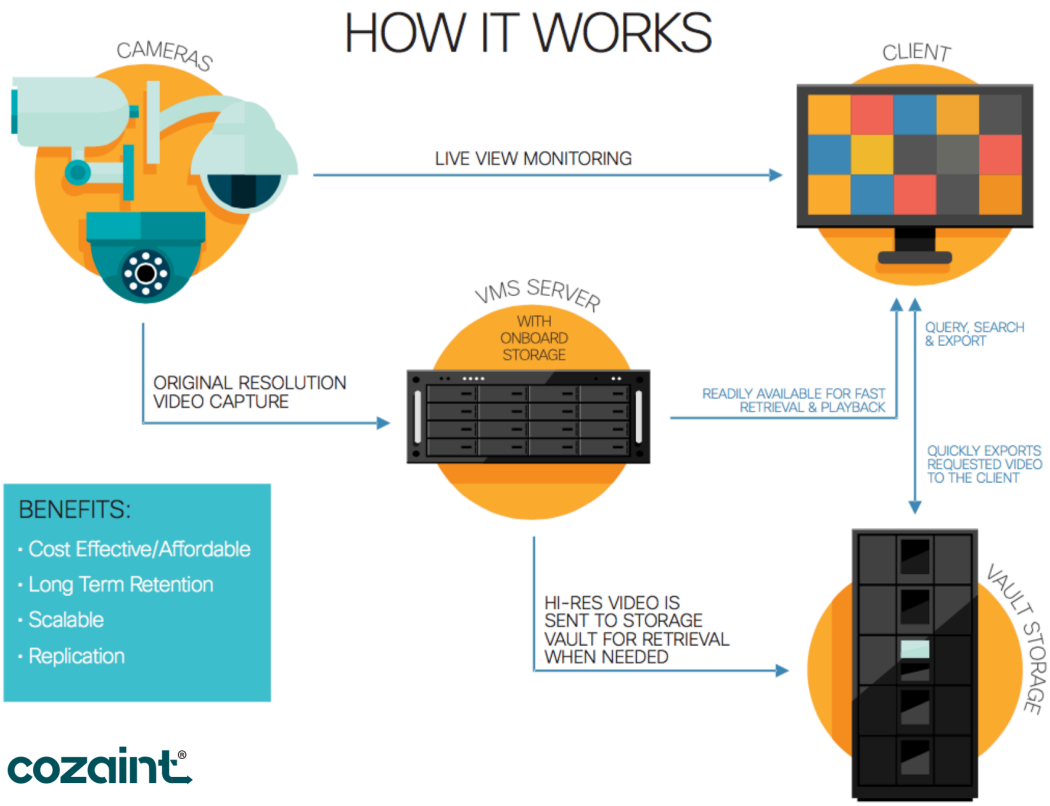
- **Compression (H.264, H.265, H.265+)** reduces storage substantially and is good practice. But as camera counts and resolutions grow, the need for more storage returns, and the cycle starts again.
- **Record-on-motion** stores video only "where there is activity." Trees, birds and flickering lights trigger false recordings, so thresholds are added, and then systems fail to record when something important happens.
- **Lower resolution, frame rate and bit rate** degrade the evidence.

# 5. The architecture of Intelligent Video Lifecycle Management

A multi-tier architecture puts each stage of a recording's life on the storage that fits it. The first edition called this "2-tier video surveillance storage." Today Cozaint calls the complete approach **Intelligent Video Lifecycle Management (IVLM)**: not just two kinds of storage, but a governed lifecycle that keeps the operator in control from the moment video is recorded to the moment it is retrieved years later.

## Three stages, one connected lifecycle

| Stage | What happens | Where the video lives |
|---|---|---|
| 01 Active video | The VMS records and plays recent video as it always has | A right-sized disk tier |
| 02 Governed retention | Retention rules decide what moves to LTO and when; new video is replicated to tape automatically in the background | LTO cartridges in the library, and offline if exported |
| 03 Connected workflow | Requested history returns to the operator through a defined workflow | Recalled from LTO to the same VMS timeline |




How a two-tier system works: cameras record to the VMS server's disk tier, video is replicated to the LTO vault, and the operator's client queries, plays back and exports any recording

## Tier 1: disk for active video

The first tier holds the most recent video, typically days to a few weeks, on high-performance disk. This is where operators review most incidents, so playback must be instant. Because it holds only a fraction of the total video, it can be a fraction of the size of a disk-only system. A common starting point is about 10% of the total retention, adjusted to how far back operators usually look.

## Tier 2: LTO for retention

The second tier holds **every** recording for the full retention period on LTO cartridges in a tape library. It is the long-term record: very low cost per terabyte, no power for cartridges at rest, decades of archive life, and the option of offline copies.

Because video is replicated to tape as it is recorded, the archive is not a backup made after the fact. It grows continuously, and the disk tier can simply roll over recent video once its copy is safely on tape.

## The VMS must understand the tiers

Here is the part most earlier attempts missed. It is not enough to attach a tape library to a recording server. The VMS, or the layer that works with it, must understand where every piece of video lives, so that:

- data moves from disk to tape according to policy, without operator effort;
- the system knows, for any camera and any moment, which tier holds the video and, on tape, which cartridge and position;
- an operator's request for older video can be answered automatically;
- the recalled video returns to the VMS for playback and export.

Historically, VMS vendors had little incentive to support LTO. Hard disk NAS and SAN storage were universal, and end customers did not yet know to ask for a tiered solution. That is the gap MARCIA fills.

## The timeline is the key

Every VMS gives operators a **timeline**: a horizontal view of recorded time for each camera. Operators move along it to the moment they need, play, pause, scrub forward and backward, and export clips for HR, police or the courts.

In an ideal multi-tier system, nothing about that changes. The operator moves the timeline to the date and time of interest:

- if the video is on disk, it plays immediately;
- if it is on LTO, the system identifies the cartridge, the library loads it into a drive, the drive seeks to the right point, and the video comes back to the timeline, typically within a few minutes;
- either way, there are no extra steps and no IT staff involved.

Properly sizing the disk tier, and setting expectations with operators, delivers an infrastructure in which any recording is available either instantly or within minutes. Most operators readily accept that, and organizations realize substantial savings over disk-only storage.

## The editor and the operator

Video editors in film and television also work on a timeline, in a non-linear editor (NLE). The two are close cousins, with two important differences:

| | Editor's timeline | Operator's timeline |
|---|---|---|
| **Length** | Finite: one production | Endless: every camera, every day |
| **Does the user know what they need?** | Yes, the editor knows the shot | Often not; the operator is searching |

Those differences explain why earlier ways of using tape failed in surveillance:

- **Pre-fetch** assumes the operator knows in advance what to restore. They usually don't; that is why they are reviewing video.
- **Backup and restore** makes tape an IT process. Where is the video restored to, and which system plays it? Every step and every ticket makes the archive less likely to be used.

The lesson: **security is not an IT process. The surveillance operator is king.** The archive must come to the operator's timeline.

## Two ways to connect a VMS and an archive

Integration between a VMS and LTO can be built through a middleware layer that translates VMS requests into storage commands, or through an application programming interface (API) that lets the two systems communicate directly. Either way, the measure of success is the same: whether the operator can ignore where the video is stored.

# 6. MARCIA and the patented workflow

Cozaint MARCIA™ Intelligent Video Lifecycle Management is the layer between the VMS and long-term storage that makes the architecture in Chapter 5 work in practice. Its patented architecture lets an operator's normal VMS playback request recall archived video from LTO and play it on the same timeline they use every day.

**U.S. Patent No. 12,755,902**, issued October 6, 2026 to Cozaint Corporation (application No. 18/065,506).

## The five steps

**Request.**
The operator moves the VMS timeline to a past date and time.

**Locate.**
MARCIA identifies which storage tier holds that video.

**Map.**
It maps the request to the exact cartridge and position.

**Recall.**
The tape library loads the cartridge and reads the video.

**Play.**
The video returns to the same VMS timeline.

| Step | What the operator sees | What MARCIA does |
|---|---|---|
| 1 Request | Moves the timeline, exactly as for recent video | Receives the playback request for that camera and time |
| 2 Locate | Nothing new | Determines whether the video is on disk or on LTO |
| 3 Map | Nothing new | Maps camera and time to the cartridge and position where the video was written |
| 4 Recall | A visible retrieval status | Directs the library to load the cartridge and reads the requested video |
| 5 Play | The video plays on the same timeline | Returns the original recording to the VMS for playback and export |

## Why this is different

Competitors can sell tape libraries. Competitors can sell hierarchical storage management (HSM) software. Competitors can build disk archives. None of those, on their own, makes LTO part of the operator's normal workflow. The patent explains why MARCIA is not simply another tape product: it is the integration that makes the storage tier invisible to the operator.

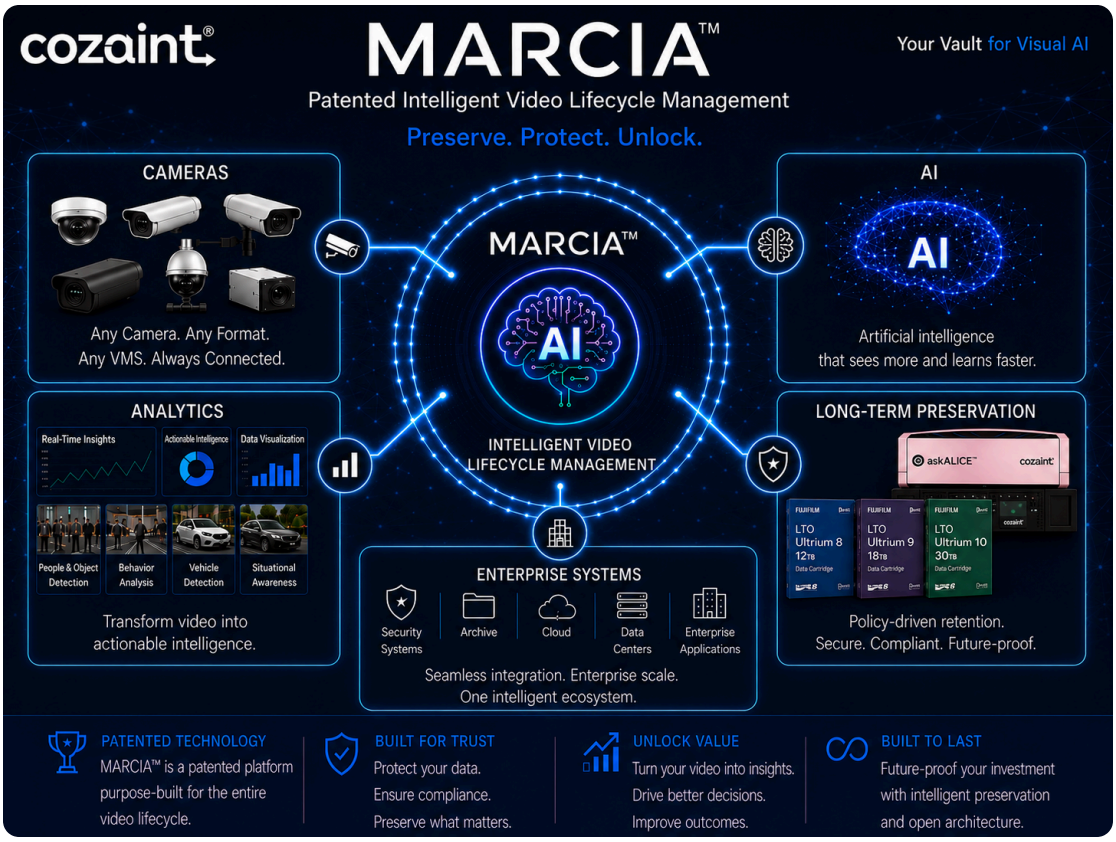
## Design choices that matter

| Choice | Why it matters |
|---|---|
| **Video for each camera kept together on tape** | Recalling a camera's history touches as few cartridges as possible |
| **At least two LTO drives** | One drive can serve a recall while another keeps writing new video |
| **Policy-driven movement** | Retention follows the organization's rules |
| **Visible request and retrieval status** | Operators know their video is on the way |
| **Open LTO format (LTFS)** | Cartridges are self-describing and readable on standard systems |
| **Offline cartridges and WORM media** | Air-gapped copies and write-once recordings where policy requires them |

> **Being precise about protection.** A cartridge is air-gapped only when it is removed from the library or otherwise disconnected; cartridges in the library are online. Recordings are protected from being overwritten when written to WORM media or governed by retention policy. Each deployment should be designed to the level of protection the organization's policy requires.

## Compatibility

MARCIA works with Milestone XProtect, Hanwha WAVE, Digital Watchdog DW Spectrum, Network Optix Nx Witness and Cozaint BOBBYvms, and is validated on HPE StoreEver and Quantum Scalar tape libraries and ATTO XstreamCORE.




MARCIA: Intelligent Video Lifecycle Management connects cameras, analytics, AI and long-term preservation

## One platform, three ways to deploy

| | MARCIA™ | askALICE™ | ALICE1™ |
|---|---|---|---|
| What it is | The Intelligent Video Lifecycle Management platform | Enterprise turnkey disk-plus-LTO system, powered by MARCIA | Compact LTO archive appliance, powered by MARCIA |
| Best for | Enterprises adding a retention tier to the VMS they run | Large sites that want one complete system | Smaller and self-contained sites |

# 7. Hollywood comes to video surveillance

Dailies. B-roll. Circle takes. For more than a decade, the film and television production business has lived with an explosion of recorded video, and has built entirely new workflows to handle it.

In the days of film, parts of movies, TV shows and commercials ended up on the cutting-room floor. Many in Media & Entertainment (M&E) point to digital cinema cameras such as RED, first conceived in 2005, as the tipping point. Once the cost of film stock disappeared, productions began to keep everything: every take might be reused in a bloopers reel, a director's cut or a sequel. One small side effect: there is a lot of extra video.

## The questions Hollywood already answered

How do directors and producers quickly review today's shoot? How do editors find a scene recorded for another production years ago and reuse it? What is the most affordable way to store all of it? Hollywood has answered each of these, and the surveillance industry faces the same questions with the same kind of data.

## Two things Hollywood got right

**Metadata.** Editors tag footage so it can be found later. "Airplane flyover of the Golden Gate Bridge at sunset" is a clip a production company can find in seconds and reuse rather than sending another crew to shoot it. In surveillance, video analytics plays the same role: people, vehicles, objects and events make years of footage searchable, and that metadata becomes worth its weight in gold.

**Multiple tiers of storage.** Just as in surveillance, production video is rarely viewed again after the first days. Hollywood keeps active footage on fast disk and archives everything to LTO. LTO meets the needs of both the studios and the insurance and bonding companies behind many productions, which have required footage captured on set or on location to be archived to LTO daily, on the same tape-based platform that banks use.

## The lesson for surveillance

Hollywood also learned that if you cannot get back to recorded video quickly and easily, there is little point in keeping it. Just like an editor, a surveillance operator cannot afford to wait or to hand the search to someone else. Operators need to scrub forward and backward through the timeline without concern for where the video is stored, then select the clip they need, directing the system to the original, untouched recording on its LTO cartridge. Done this way, multi-tier storage also delivers the chain of custody needed to submit video as evidence.

From a cost standpoint, the comparison is stark. Compare the acquisition cost of terabytes and petabytes of spinning NAS and SAN storage with an LTO library complete with robotics. Add the three-to-five-year life of spinning disks against decades of LTO archive life. Add the electricity and cooling for all that spinning disk, 24/7, year-round.

When contemplating a 21st-century surveillance video management solution, a disk-only approach is like navigating before anyone knew the world was round. There are better ways to reach our destination. We all want to store as much video as we can, at the highest resolution and the highest frame rates, as affordably as possible. So why not learn from our colleagues in Hollywood? Embracing the sequential structure of video and LTO data tape delivers the ideal storage solution, and done properly, LTO multi-tier storage is surveillance video's "killer app."

Thanks, Hollywood.

# 8. What about the cloud?

Cloud storage is often presented as the easy answer for video retention: flexible, scalable and cheap to start. For long-term surveillance retention, the costs and limits are rarely discussed up front.

## Bandwidth

Every camera's video must be uploaded continuously. A 1080p camera at 30 frames per second can need on the order of 20 Mbps, and a 4K camera at 60 frames per second around 100 Mbps. Sites with many high-definition cameras need dedicated high-speed connections, often fiber, which carry significant monthly costs and installation fees, and even a 1 Gbps connection may not be enough. Any outage in connectivity affects the recording system.

## Recurring storage fees

Cloud providers charge every month for every terabyte stored. The calculator behind Chapter 10 uses the average standard, pay-as-you-go price of four leading providers, about $16.50 to $17 per TB per month at archive scale. Keeping one petabyte for a year at that price is roughly $200,000, and the bill never stops while the video is kept.

## Fees to get your video back

Many providers charge egress or retrieval fees to download data, along with request charges, minimum storage periods and early-deletion fees. Investigations that need to pull large amounts of video can trigger significant, unpredictable bills.

## Ownership, control and security

Organizations must check who controls their data, how it can be accessed, and what rights they retain to access, download and delete it. Video is sensitive, and moving it off-site means relying on another organization's security practices, often with less visibility into who accessed what.

## And still the workflow problem

Even when the costs are acceptable, cloud archives often require operators to use a separate portal rather than the VMS timeline they use every day.

| | Disk only | Cloud | MARCIA (disk + LTO) |
|---|---|---|---|
| Immediate access to recent video | Yes | Yes | Yes, recent tier on disk |
| Multi-year economics | Poor | Variable, recurring fees | Strong |

| | Disk only | Cloud | MARCIA (disk + LTO) |
|---|---|---|---|
| Power for archived video | Continuous | Provider's | None for cartridges at rest |
| Depends on internet bandwidth | No | Yes | No |
| Offline (air-gapped) copy | Limited | No | Yes, cartridges can be stored offline |
| Operator workflow | VMS timeline | Often a separate portal | Same VMS timeline |
| Scaling to petabytes | Expensive | Expensive, recurring | Designed for it |

# 9. Sizing a system

Sizing a lifecycle architecture starts with the same question as any surveillance storage design: how much video will be recorded, and for how long must it be kept? The difference is what happens next. Instead of buying disk for the whole answer, you size a small disk tier for active video and an LTO tier for the full retention period.

## What determines the amount of video

| Factor | Effect |
|---|---|
| **Number of cameras** | More cameras, more video |
| **Resolution** | The single biggest factor: more pixels per frame, more data |
| **Frame rate** | More frames per second, more data |
| **Motion** | Busy scenes compress less than quiet ones |
| **Compression** | H.264 and H.265 reduce the bit rate substantially |
| **Retention** | How many days of video must be kept, which a lifecycle architecture lets you choose rather than settle for |

## From bit rate to terabytes

The most practical way to size storage is from each camera's recording bit rate, the amount of data it records per second after compression:

- **Data per camera per day** = bit rate (kilobits per second) ÷ 8 × 86,400 seconds
- **Total storage** = data per camera per day × number of cameras × retention days

Cozaint's storage calculator at cozaint.com/assessment estimates the bit rate from resolution, frame rate and motion, or lets you enter it directly, and does the arithmetic.

## Compression

H.264 (also called AVC) divides each frame into blocks and uses motion estimation, transform coding and entropy coding to remove redundancy between and within frames. It produces far smaller files than older standards such as MPEG-2 while keeping quality high. H.265 (HEVC) compresses further at the cost of more processing power. Motion JPEG compresses each frame separately and produces much larger files. Compression is good practice, but it does not change the underlying trend: as cameras and resolutions grow, storage grows with them.

## Frame rate: why more is often better

The human eye does not see in frames, but studies suggest people can distinguish separate images at around 60 frames per second or more under some conditions. Higher frame rates capture motion with greater temporal detail, and when played back in slow motion they reveal what normal speed hides.

Casino surveillance is a good example. In games such as blackjack, slow-motion review helps surveillance teams spot card switching, chip manipulation, dealer collusion or improper shuffling. A player swapping a low card for a high one from a sleeve may be visible only frame by frame. The only reason to record at less than the camera's full frame rate is storage cost. Remove that constraint, and the argument favors recording at 30 frames per second or more.

## A worked example

| Input | Value |
|---|---|
| Cameras | 50 |
| Resolution | 1080p |
| Average motion | 30% |
| Frame rate | 30 fps |
| Compression | H.264, about 3,110 kbps per camera |
| Retention | 180 days |
| **Total video** | **about 281.5 TB** |

**Disk-only design.** A traditional single-tier design needs enough RAID-protected disk for all 281.5 TB: for example, a 32-bay server populated with large drives in RAID.

**Lifecycle design.** Working with the operators, the disk tier is set at about 10% of the total, roughly 28 TB, rounded up to 30 TB to allow for RAID protection and overhead. That fits in a small four-bay NAS-class server. The LTO tier is a single 2U library with 24 slots, using only 16 LTO-9 cartridges for the full 180 days.

Comparing equipment and energy costs, the lifecycle design came to about **45% lower overall cost** than the traditional design, and it scales by adding cartridges as retention grows.

## Tape drives and throughput

A single LTO drive can keep up with many concurrent camera streams. Best practice is at least two drives, so one can serve an operator's recall while recording continues to be written. Larger installations use larger libraries with more drives and hundreds of slots. The calculator estimates the drive count from the daily volume of video and the time needed to write each cartridge.

# 10. The economics

Most surveillance storage decisions are made on purchase price, but most of the money is spent afterwards: power and cooling around the clock, replacement drives, and above all the cost of keeping more video every year.

The companion paper *The Economics of Long-Term Video Retention* compares three ways to keep **one year** of video, using the same price model as the storage calculator at cozaint.com/assessment (prices checked November 1, 2026):

- **Disk-only:** the average price per terabyte of six complete video surveillance storage systems, with RAID 6, volume price breaks, power and cooling.
- **Cloud:** the average standard storage price of four cloud providers.
- **MARCIA:** a 10% disk tier plus LTO-9, with Cozaint products at full list price.

## Five-year cost

| Archive size | MARCIA (disk tier + LTO) | Disk-only system | Cloud storage | MARCIA saves vs disk | MARCIA saves vs cloud |
|---|---|---|---|---|---|
| 500 TB | $117,762 | $153,229 | $508,680 | 23% | 77% |
| 1 PB | $145,062 | $259,628 | $1,003,080 | 44% | 86% |
| 2 PB | $207,050 | $496,739 | $1,991,880 | 58% | 90% |
| 5 PB | $373,530 | $1,191,981 | $4,958,280 | 69% | 92% |

## What the numbers show

1. **Scale changes the answer.** The tape tier carries fixed costs (a library, drives, software), so it pays off as the archive grows. Above 1 PB, MARCIA's advantage is large and keeps growing.
2. **Retention changes the answer.** Disk cost grows in direct proportion to the video kept. With MARCIA, keeping more video mostly means adding cartridges.
3. **Cloud is the most expensive way to keep a year of video,** before counting upload bandwidth and retrieval fees.
4. **The model leans in the competitors' favor.** Disk and cloud are at discounted or published prices, Cozaint at full list price, and the disk figures leave out rack space, support contracts and full system replacement after five years.

Across a range of earlier studies and deployments, organizations have typically realized **40% to 60% savings** over traditional surveillance storage with a two-tier disk-and-LTO design, and more as archives reach petabyte scale.

## Energy and sustainability

LTO-based storage is a leading “green” choice for surveillance retention. Cartridges at rest draw no power and need no cooling, and studies cited by the tape industry put tape’s long-term energy use about 87% below an equivalent capacity of disk, with about 97% less CO2e over the product lifecycle. That helps organizations meet ESG goals as well as budgets.

# 11. Real-world examples

The two-tier disk-and-LTO architecture has been applied in large and demanding surveillance environments.

## New York City Police Department

The NYPD operates one of the largest video surveillance systems in the world, with more than 18,000 cameras deployed across the city. It implemented a two-tier storage solution with LTO technology to manage its video: an active tier of high-performance SSD and HDD storage for short-term retention and a cold tier of LTO tape for long-term retention. The NYPD estimated that the two-tier architecture reduced the cost of storage by **more than 70%** compared to traditional storage solutions.

## NASA Kennedy Space Center

NASA's Kennedy Space Center implemented a two-tier video surveillance storage solution using LTO technology to manage the large amounts of video generated by its facilities. Short-term video is kept on high-performance storage, and long-term retention on LTO tapes, which are stored off-site for disaster recovery. The architecture enabled the center to store and manage its video efficiently and cost-effectively.

## University of Oregon

The University of Oregon implemented a two-tier solution with LTO technology for its campus-wide surveillance system, with an active tier for short-term video and LTO for long-term retention. The university estimated that the architecture reduced the cost of storage by **more than 50%** compared to traditional storage solutions.

## City of Oceanside, California

At the City of Oceanside's San Luis Rey Wastewater Treatment Plant, California records rules require certain government records, including video, to be kept for one year. More than 30 high-definition (1080p) cameras record 24/7, which comes to about 548 TB for a year. A disk-only system was too expensive for the City's budget.

The City deployed Cozaint's askALICE system, the disk-and-LTO architecture that MARCIA is built on. Using the FUJIFILM video surveillance retention cost calculator with the plant's own requirements, the two-tier design came to **$31,515 over five years against $68,506 for disk-only, a 54% saving.** Operators play back any recording from any point in the year themselves, with no help from IT.

> “We needed to upgrade our security surveillance system at our water treatment plant, and since we have worked with several different systems, we knew what we liked and what we didn’t like. It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint’s askALICE has been an ideal fit for our surveillance recording needs and our budget.”
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

The full story is in the case study *The Problem MARCIA Was Built to Solve*.

## Return on investment

Organizations that adopt a lifecycle architecture can expect a return on their investment within a few years, and the savings grow as video accumulates. The scalability and longevity of LTO keep delivering value long after the purchase.

# 12. The physical architecture

A lifecycle architecture uses standard, proven components. This chapter summarizes how they connect; the *How It All Connects* guide shows each step.

## The basic connection

| Component | Role |
|---|---|
| **Server** | A standard rackmount server runs the VMS recording, the disk tier and MARCIA |
| **SAS host bus adapter (HBA)** | A card installed in a PCIe slot of the server that connects it to external storage |
| **External SAS cable** | Runs from the HBA's port to the library (keyed connectors ensure correct alignment) |
| **LTO library** | Holds the drives and cartridges; MARCIA directs it to write and recall video |

## Installing a SAS connection

1. Power off the server and the library.
2. Install the SAS HBA in an available PCIe slot and secure it.
3. Connect the SAS cable from the HBA to the library's SAS port.
4. Power on the library first and let it become fully ready, then power on the server.
5. Install the HBA's drivers for the server's operating system, and reboot if needed.
6. Confirm the server sees the library and its drives, then initialize and configure the library according to the manufacturer's instructions.

## Fibre Channel

Larger installations may connect through Fibre Channel (FC) HBAs and, where several devices share the connection, an FC switch, configured with zoning according to the manufacturer's guidelines. FC adds performance and redundancy for large libraries.

## More than one library

Several libraries can connect to one server for more capacity or redundancy, provided the server has enough PCIe slots, processing, memory and bandwidth, and its operating system supports multiple HBAs. A multi-connector "hydra" cable can consolidate several SAS connections and simplify cabling.

## Built to scale

The same architecture runs from a compact 2U library with a single drive and two dozen slots to stacked, modular libraries and full 42U racks, and on to enterprise libraries with dozens or hundreds of drives and thousands of cartridges. Examples include the IBM TS4500, HPE StoreEver MSL6480, Quantum Scalar i6000 and Spectra Logic T-Finity families. Most libraries grow by adding modules, drives and cartridges, so a system can start small and expand with retention.

# 13. Best practices and getting started

## Design

1. **Record at the quality the cameras were bought for.** Change the storage, not the evidence.
2. **Size the disk tier for how far back operators usually look,** typically days to a few weeks, about 10% of total retention as a starting point.
3. **Put the full retention period on LTO,** with room to grow by adding cartridges.
4. **Use at least two tape drives,** so recalls and recording never compete.
5. **Use RAID on the disk tier** to protect recent video and keep performance high.
6. **Avoid SMR hard drives** in recording systems.
7. **Decide your protection policy:** offline cartridges for air-gapped copies, WORM media where recordings must not be overwritten, and encryption where needed.

## Operation

- **Keep operators in the VMS.** The test of the design is whether they can ignore where the video is stored.
- **Set expectations:** recent video is instant; older video on tape typically returns within minutes.
- **Monitor the system:** library health, drive cleaning and calibration, cartridge use and retention policy.
- **Train staff** on playback, export and evidence handling.

## Getting started

1. **Start with your numbers.** Use the storage calculator at cozaint.com/assessment to size your system and compare MARCIA, disk and cloud, and download the PDF report.
2. **Evaluate what you have.** Review current capacity, retention, playback experience and cost, including power and replacement.
3. **Start small if you like.** A pilot on one site or one group of cameras proves the workflow with minimal risk.
4. **Work with an experienced partner.** Cozaint and its reseller partners design, install and support lifecycle architectures for enterprise surveillance.

## The benefits, in summary

| Benefit | What it means |
|---|---|
| Cost-effective | Substantially lower lifecycle cost than disk-only or cloud, growing with scale |

| Benefit | What it means |
|---|---|
| **Scalable** | Grow retention by adding cartridges; grow libraries by adding modules |
| **Fast where it matters** | Recent video instantly from disk; older video within minutes |
| **High capacity** | 30 TB per LTO-10 cartridge, up to 40 TB with the newest cartridges, and a roadmap to 576 TB |
| **Long-lived** | Media rated for decades in archive |
| **Energy efficient** | No power for cartridges at rest |
| **Secure** | Offline copies, WORM media and encryption |
| **Operator-friendly** | The patented MARCIA workflow keeps every recording on the same VMS timeline |

**Media & Entertainment proved LTO for video. MARCIA makes it practical for surveillance. Patented VMS integration makes the storage tier invisible to the operator.**

**Talk to Cozaint.** Request a MARCIA assessment at **cozaint.com/marcia**, run your numbers at **cozaint.com/assessment**, or contact **info@cozaint.com** · **760-975-8000**.

## About the author and Cozaint

Jay Jason Bartlett is the CEO of Cozaint Corporation and the author of the first edition of this book (2023).

Founded in 2018, Cozaint Corporation is a service-disabled veteran-owned business that develops video surveillance and storage solutions, including MARCIA™ Intelligent Video Lifecycle Management, askALICE™, ALICE1™ and BOBBYvms. Cozaint is headquartered in Palmetto Bay, Florida, with an office in San Diego, California.

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